

Mixed Reality 3D User Interfaces for Volumetric Field Analysis in Stadium-Scale Building

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Figure 1: A vision-based mixed reality 3DUI for analyzing temperature fields in large buildings. It overlays CFD temperature data onto the real-world-scale BIM model, using sparse numeric anchors and rising particles to represent values and convection

ABSTRACT

Understanding volumetric environmental fields (e.g., temperature and airflow) is essential for operating large public buildings, yet conventional 2D visualizations provide limited spatial insight. This poster presents a mixed-reality immersive analytics system that integrates CFD simulations with BIM and visualizes scalar and vector fields at true spatial scale. Scalars are encoded using perceptually guided colormaps and convection-inspired particles, while airflow is depicted with streamlines and semantics-aware filtering. Preliminary results from a controlled study ($n = 16$) show faster task completion and higher accuracy in MR compared to a desktop baseline, and clear advantages of streamline over arrow encodings. These findings suggest that MR-based 3D user interfaces can support more efficient and intuitive large-scale environmental analysis.

Index terms: Virtual Reality, Mixed Reality, Immersive analytics, Flow visualization, 3D user interfaces, Building information modeling.

1 INTRODUCTION

Recent advances in virtual and mixed reality (VR/MR) and available head-mounted displays (HMDs) support immersive analytics for situated data analysis. Mixed-reality HMDs align Building Information Modeling (BIM) data with the physical environment, enabling in situ reasoning at true spatial scale [1-3].

In building operations, environmental fields such as temperature, humidity, and airflow are critical to indoor quality and energy performance. These data are inherently volumetric and commonly

derived from computational fluid dynamics (CFD), yet they are still predominantly analyzed using desktop-based 2D visualizations. Such approaches struggle to convey spatial relationships at scale or to integrate multiple physical variables coherently. These limitations are amplified in large public buildings.

To address these issues, we present a mixed-reality immersive analytics system for volumetric environmental fields in stadium-scale buildings. The system integrates CFD data with BIM geometry and supports real-scale exploration using perceptually guided scalar encodings, streamline-based airflow visualization, cross-sectional views, and semantics-aware filtering. An initial user study examines whether immersive visualization improves task efficiency and accuracy compared to a desktop baseline, and whether streamlines outperform arrow glyphs for airflow understanding [3, 4].

2 MATERIALS AND METHODS

We leverage 3D, real-scale spatial context to support reasoning about volumetric environmental fields and their relationship to the built envelope. The system operates in both VR mode (for controlled laboratory evaluation) and MR mode (for in-situ analysis), sharing the same data pipeline and interaction design.

2.1 Data Sources and Spatial Registration

The study uses CFD simulations of a stadium-scale public arena under event-time operating conditions. Simulated scalar fields (temperature, humidity) and vector fields (airflow) are aligned with a BIM model used solely for spatial registration and occlusion handling. To enable efficient rendering, irregular CFD samples are packed into a regular 3D lattice aligned with the BIM geometry, packed into 3D textures (scalars: single-channel; vectors: RGB).

2.2 Visual Encodings

Scalar fields (temperature, humidity). Exact values are conveyed using sparse numeric anchors placed on a controllable 3D lattice, rather than dense text overlays. Anchors scale with distance and

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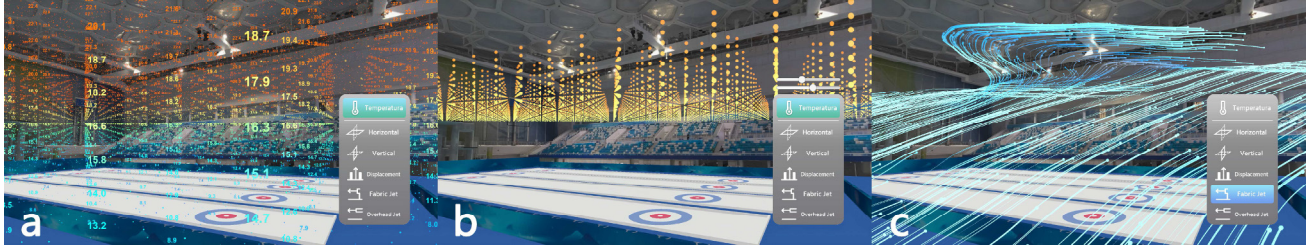


Figure 2: The visualization of physical environmental fields

fade under occlusion to reduce clutter. Values are mapped to a perceptually uniform colormap to preserve ordinal judgements. To support intuitive perception of vertical gradients, a low-density field of convection-inspired particle s rises slowly through the volume, with particle color sampled from the scalar field.

Vector fields (airflow). Airflow is visualized using strip-based streamlines. Trail length encodes local speed, while color and brightness redundantly encode magnitude. To manage visual density at stadium scale, an occupancy grid adaptively throttles particle seeding. Two seeding strategies are supported (see Section 2.3).

2.3 3D User Interface and Interaction

Users interact through a lightweight 3D user interface designed specifically for this system. A panel on the right side provides one-tap toggles for controlling the display of environmental fields. For the vector field, horizontal and vertical cross-sectional planes can be enabled, with controls for their position and thickness. When a cross-section is active, vector seeding is constrained to that slice to improve clarity. Semantics-aware filters control particle seeding locations to reduce clutter and aid interpretation. Figure 2 shows the complete workflow of this MR system.

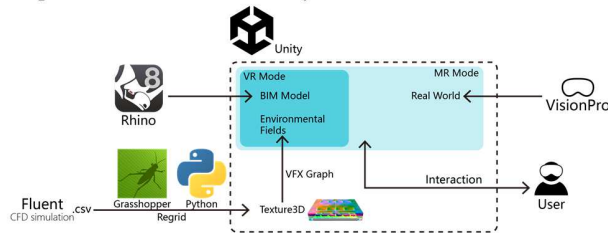


Figure 3: Process of the MR system.

3 USER STUDY AND RESULTS

3.1 Participants and Procedure

Sixteen senior undergraduate architecture students (8 female, 8 male) with prior building-physics knowledge participated in the study. After a brief introduction to the MR HMD, participants completed two task sets.

In the temperature task, participants identified temperature values at specified target locations using both a desktop visualization and the MR system. In the airflow task, participants analyzed airflow from two outlet types using either arrow glyphs or streamline-based encodings in MR, answering both direct observation and inference questions. Task order was balanced across participants to mitigate learning effects.

3.2 Results

For temperature analysis, participants completed tasks significantly faster in MR ($M = 18.5$ s) than on desktop ($M = 30.1$ s), $t(15) = 3.34$, $p = 0.004$ (Figure 4).

For airflow analysis, streamline visualization resulted in faster task completion ($M = 46.4$ s) than arrow glyphs ($M = 69.7$ s), $t(15) = 5.26$, $p < 0.001$. Streamlines also yielded higher accuracy (87.5%) than arrows (65.6%), $t(15) = -2.67$, $p = 0.017$.

These results suggest that immersive, real-scale visualization improves efficiency and accuracy in volumetric environmental analysis compared to desktop views. For airflow understanding, streamline-based encodings better convey combined direction and magnitude information, aligning with users' intuitive mental models of flow.

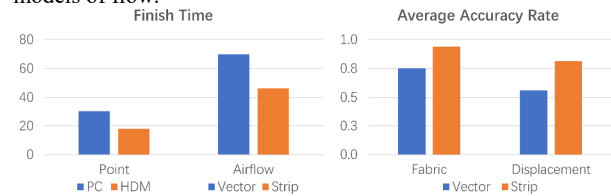


Figure 4: Results of the task.

4 DISCUSSION AND FUTURE WORK

We presented a mixed-reality immersive analytics system for visualizing CFD-derived environmental fields in large buildings at true scale. By integrating BIM geometry with perceptual scalar and vector encodings, the system facilitates analysis of temperature, humidity, and airflow. Results from a controlled study show that immersive visualization improves task efficiency and accuracy over desktop views, and that streamline-based airflow representations outperform arrow glyphs, aligning with models of continuous flow.

While this study provides initial validation, limitations exist: the system uses precomputed CFD data and was evaluated with architecture students. Future work will integrate real-time sensor data and predictive models, and involve professionals in real-world settings. Overall, this work demonstrates the potential of MR-based immersive analytics to support decision-making in complex, large-scale building environments.

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